BIOMED SCI 552:

STATISTICAL THINKING

LECTURE 4: EXPLORATORY DATA ANALYSIS AND HYPOTHESIS TESTING

QUESTIONS FROM LAST CLASS?

THE PROBLEM SET

- People found this one a little more challenging
- That's a good thing
  - There is, on occasion, virtue in being a little stuck
- Remember: If you show your work and I can follow your logic, partial credit can be had



PROBLEM 1

- This is just an extension of the Bayes' Theorem problem we covered in class

$$P(Disease|Test = Positive) = \frac{P(Test = Positive|Disease)P(Disease)}{P(Test = Positive)}$$

$$P(Disease|Test = Positive) = \frac{0.98 * 0.0133}{\big((1 - 0.0133) * 0.05\big) + (0.0133 * 0.98)}$$

$$P(Disease|Test = Positive) = 0.209$$

QUESTION 2

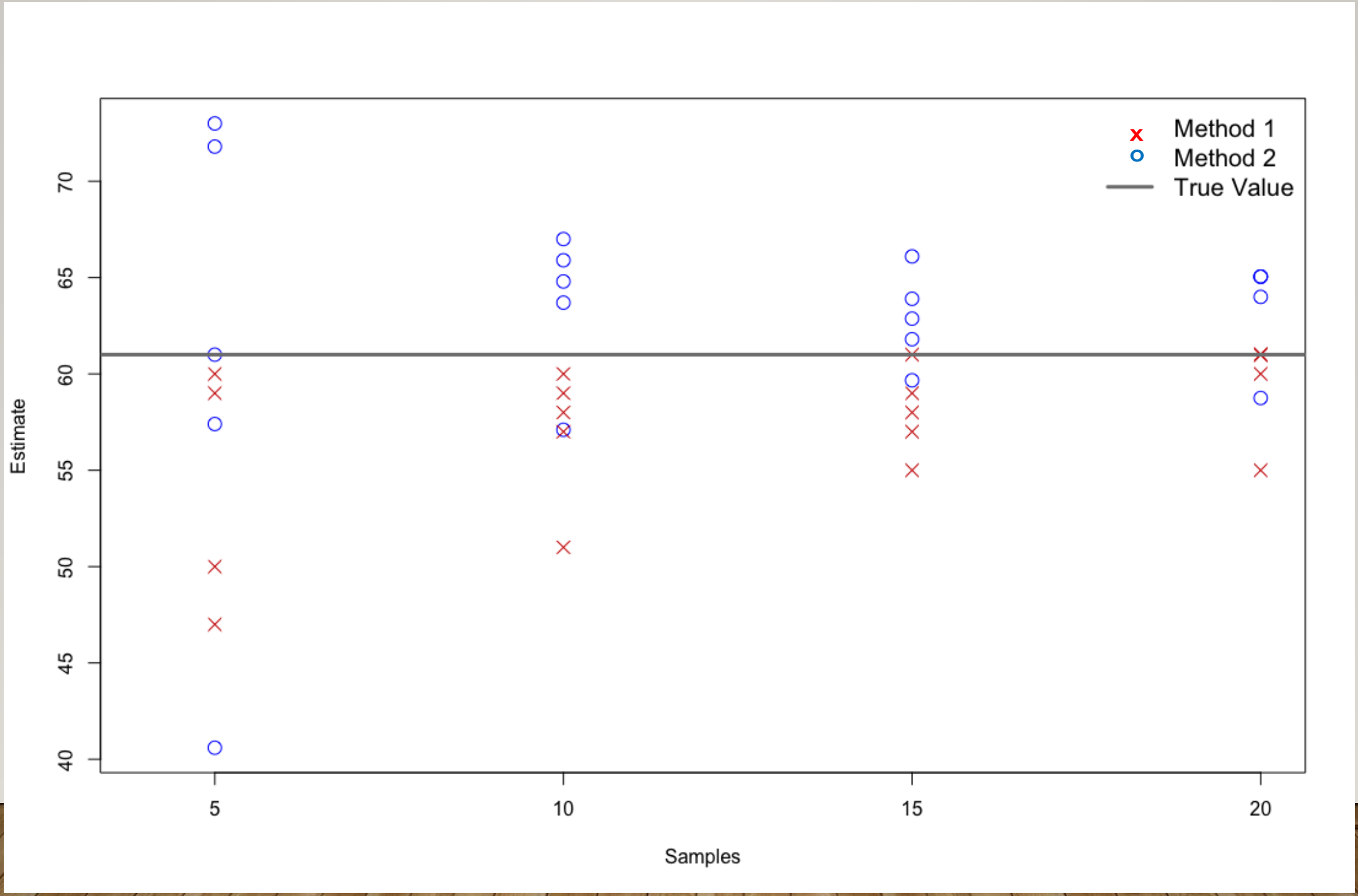


MY TAKE ON QUESTION 2

- What is the cost of the test?
- What is the consequence of not knowing vs. knowing?
- What interventions are the result of the test, and are these themselves harmful or expensive?

QUESTION 3

- This was mostly an opportunity to explore this
- There's not necessarily *a* way to answer this question


Method 1
Method 2
True Value
Estimate
40
45
50
55
60
65
70
5
10
15
20
Samples

QUESTION 4

- I've been given a problem with a probability of success (1/6) and a number of trials (20) – this feels like a binomial distribution problem to me
- It's 6 *or more* times, so I don't just need the probability of 6, but of 6 to 20
- This seems laborious to me
- The probability of 0, 1, 2, 3, 4 and 5 is one minus the probability I want, and involves way less calculation

CALCULATIONS

$$f(s) = \frac{n!}{s!\,(n-s)!} p^s (1-p)^{n-s}$$

Or…

```
dbinom(s,size,prob,log=FALSE)
```

S	P(S)	Cumulative Probability
0	0.02606	0.02606
1	0.10428	0.13034
2	0.19818	0.32852
3	0.23787	0.56638
4	0.20224	0.76862
5	0.12946	0.89808

TAKE THE BET?

- P(Winning) = 1-0.89808 = 0.10192
- That's an unlikely victory. But the payout is decent. On average, we expect to make $100*0.10192 = $10.192 on the bet. Let's round that to $10.19
- On the other hand, on average, we expect to *lose* 0.89808*$20 or $17.96 on the bet
- Maybe find some other activity to occupy your time

IS BUYING A SINGLE LOTTERY TICKET RATIONAL?











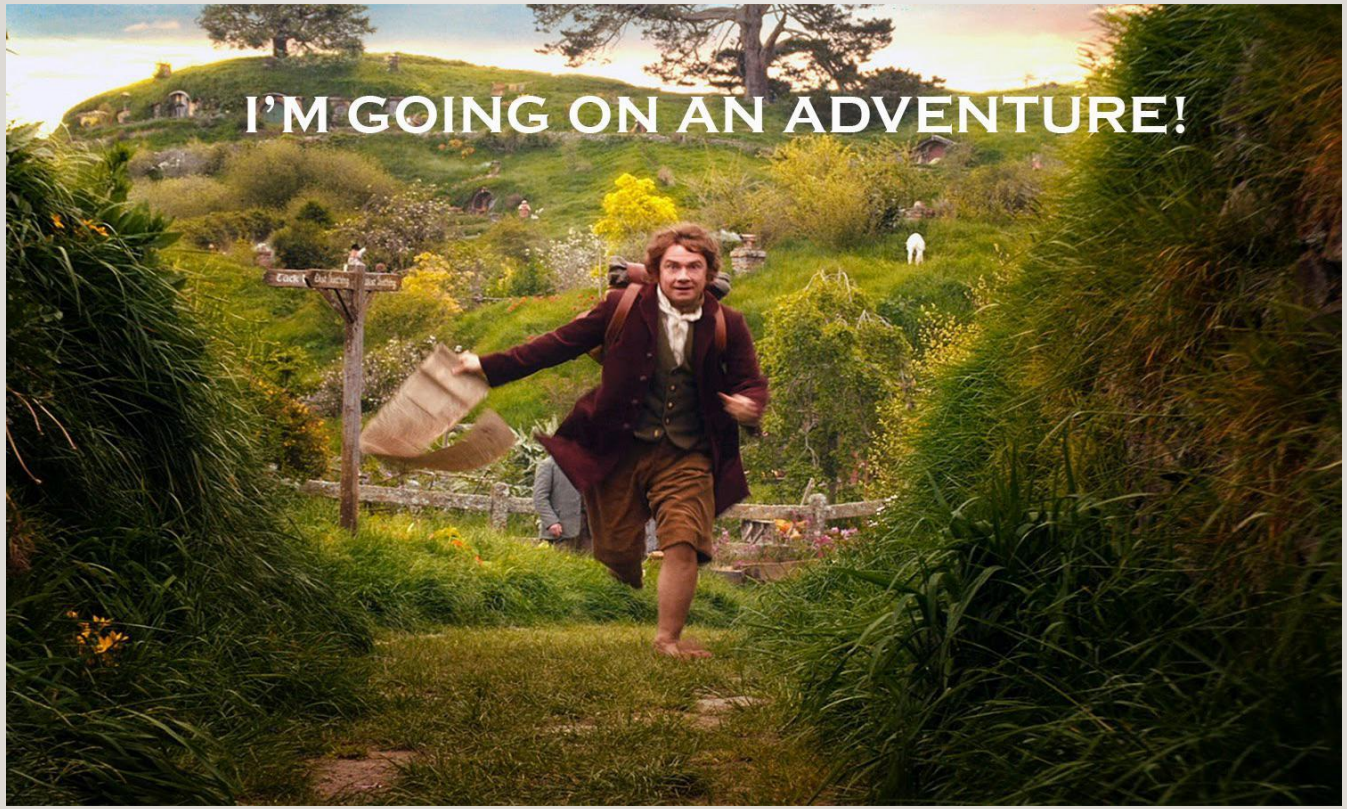
MAYBE…

- One lottery ticket is a rational purchase *if* the entertainment value of the fantasy of winning exceeds the cost of the ticket
- Expected Value of a Lottery Ticket:
- Cost = Ticket Price
- Value = Fantasy + A Potentially Large Number * Basically Zero = Fantasy

- **Two** lottery tickets doubles the price, for no increased value (you haven't increased the value of the fantasy, and the expected monetary reward is still, essentially, zero)

QUESTIONS?

EXPLORATORY DATA ANALYSIS



WHAT IS EXPLORATORY DATA ANALYSIS

- First, we're going to call it EDA for now to keep things short
- EDA is the process of…well…exploring your data
- EDA is an exploration of the characteristics of your data, often using data visualization, to help you identify patterns, spot errors, and suggest directions to take an analysis in
- It's very helpful for assessing your assumptions
- Essentially invented by John Tukey in 1970

ELEMENTS OF EDA

- Univariate, non-graphical techniques
- Univariate graphical techniques
- Multivariate, non-graphical techniques
- Multivariate, graphical techniques

UNIVARIATE TECHNIQUES

- These techniques are primarily about simply describing a single variable, to help you understand it
- What sort of distribution might it follow?
- Are there outliers or extreme values?
  - Are these in error, or are they simply extreme values?
  - I once found a 2000-year-old man in what was theoretically a clean data set
  - I also found some test data accidentally left in, because it looked funny

FIVE NUMBER SUMMARY OF A VARIABLE

- Two Extremes:
  - Minimum and Maximum
- The Median
- The Quartiles (aka the 25th and 75th percentiles)
- Why these and not the mean and standard deviation?

AN EXPERIMENT

- I give you two samples with the following information:

Sample 1 Mean	Sample 1 SD	Sample 2 Mean	Sample 2 SD
30.095	4.99	30.1202	8.99

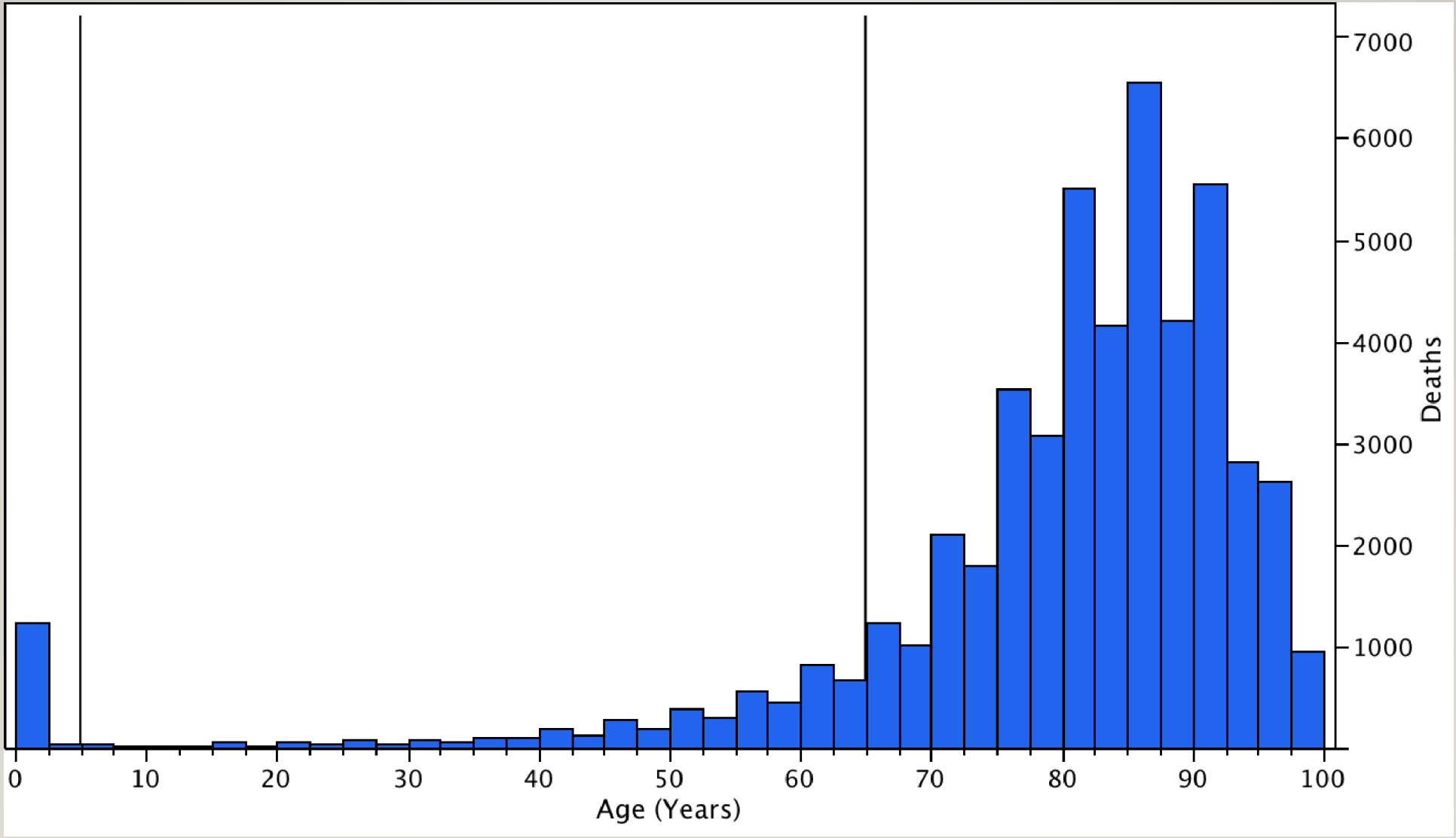
- Are these different samples? What's wrong? It's fairly hard to tell…
- Lets look at a five number summary, using the fivenum() function in R…

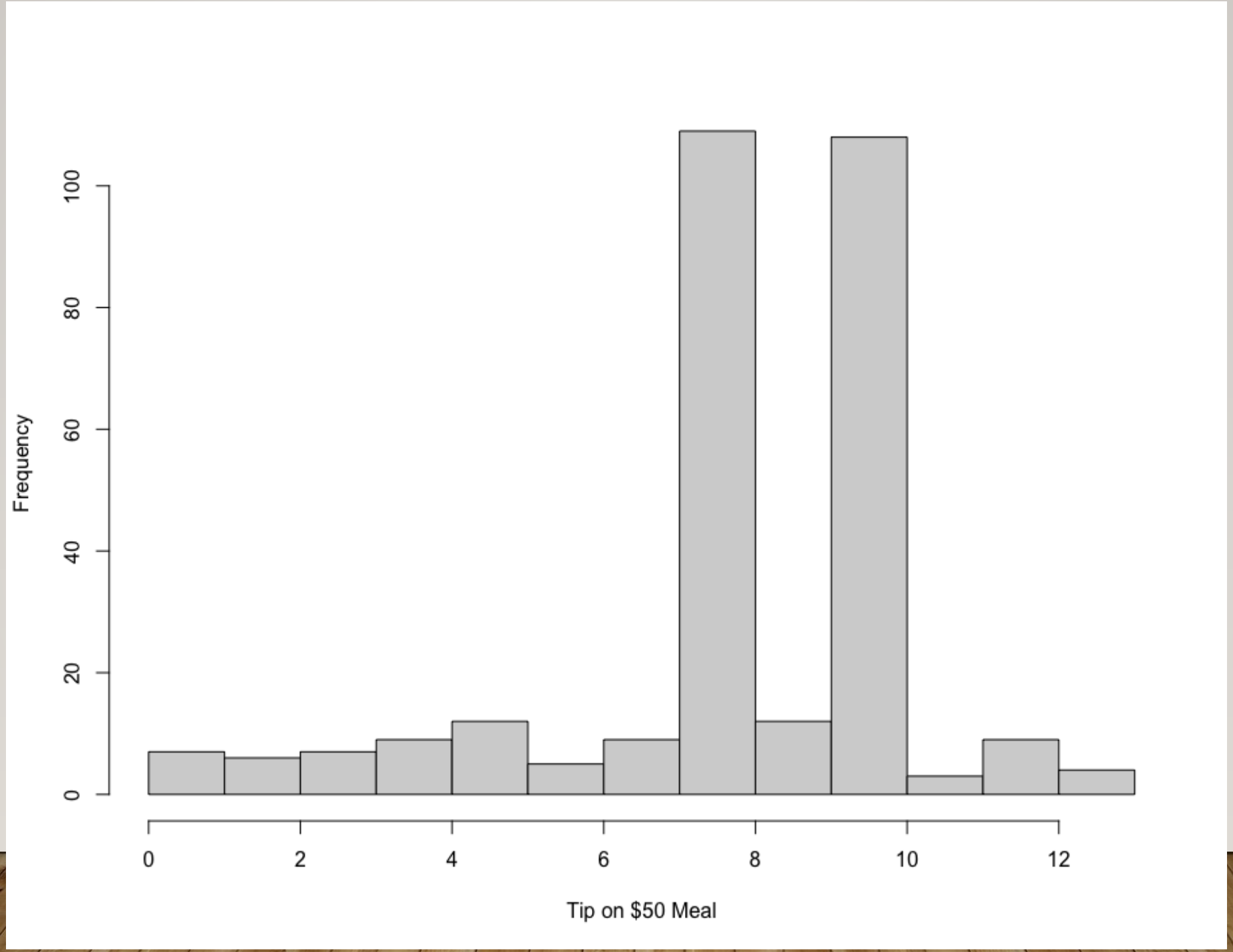
Sample	Min	25th Percentile	Median	75th Percentile	Max
1	10.87	26.68	30.11	33.42	49.22
2	11.60	26.65	30.03	33.45	778.0

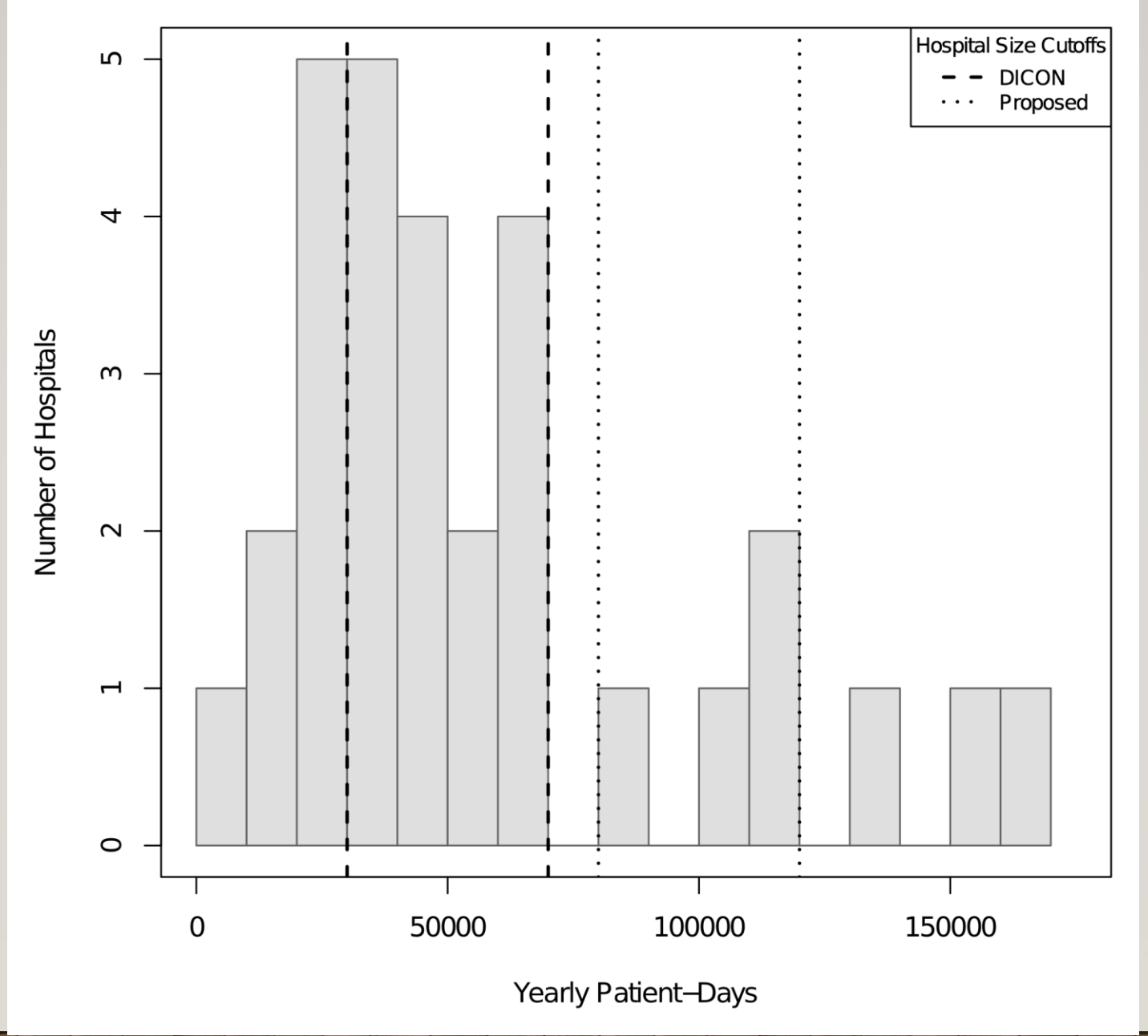
The problem is now obvious

VISUALIZING A SINGLE VARIABLE

- You can also visualize a variable using a histogram, density plot, or other methods
- This can show you things like break points in your data, natural cut offs, etc.
  - This is often very important if you want to categorize a continuous variable – it's often helpful to make sure you're not cutting a logical grouping in half
- This can also help you detect if there are thresholds in your data you were (or were not) expecting
  - Do values of a lab assay suddenly truncate?
  - Do you not enroll anyone of a particular value or lower (or higher) even though you know they exist?

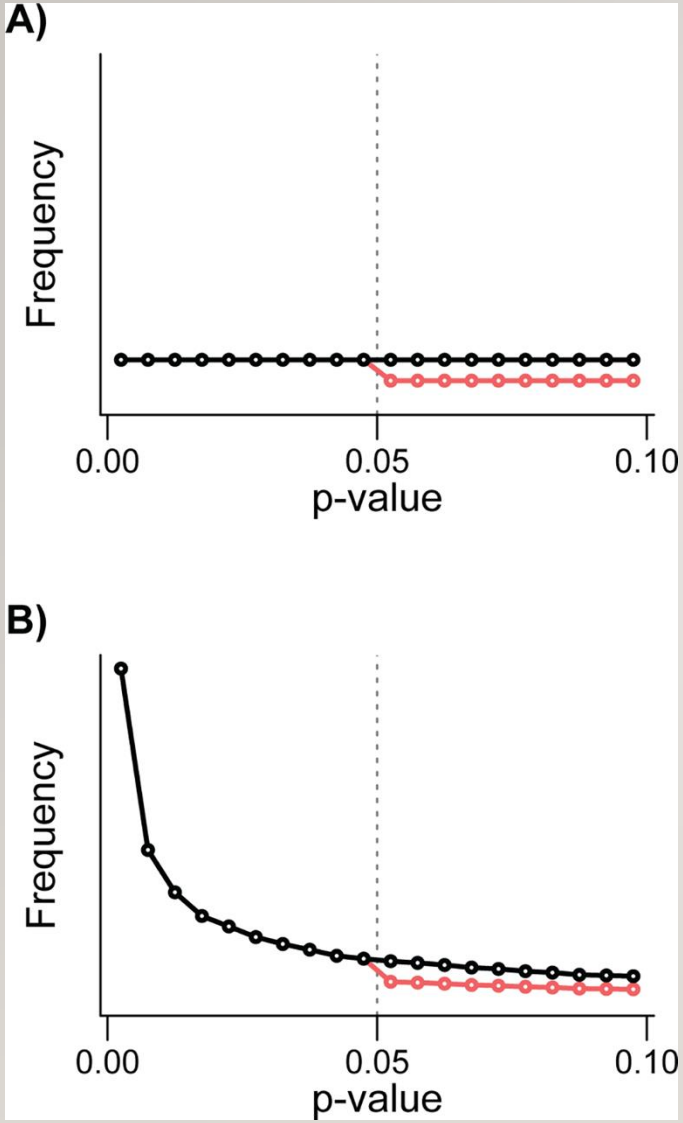

7000
6000
5000
4000
3000
2000
1000
Deaths
0
10
20
30
40
50
60
70
80
90
100
Age (Years)


Frequency
0
20
40
60
80
100
0
2
4
6
8
10
12
Tip on $50 Meal

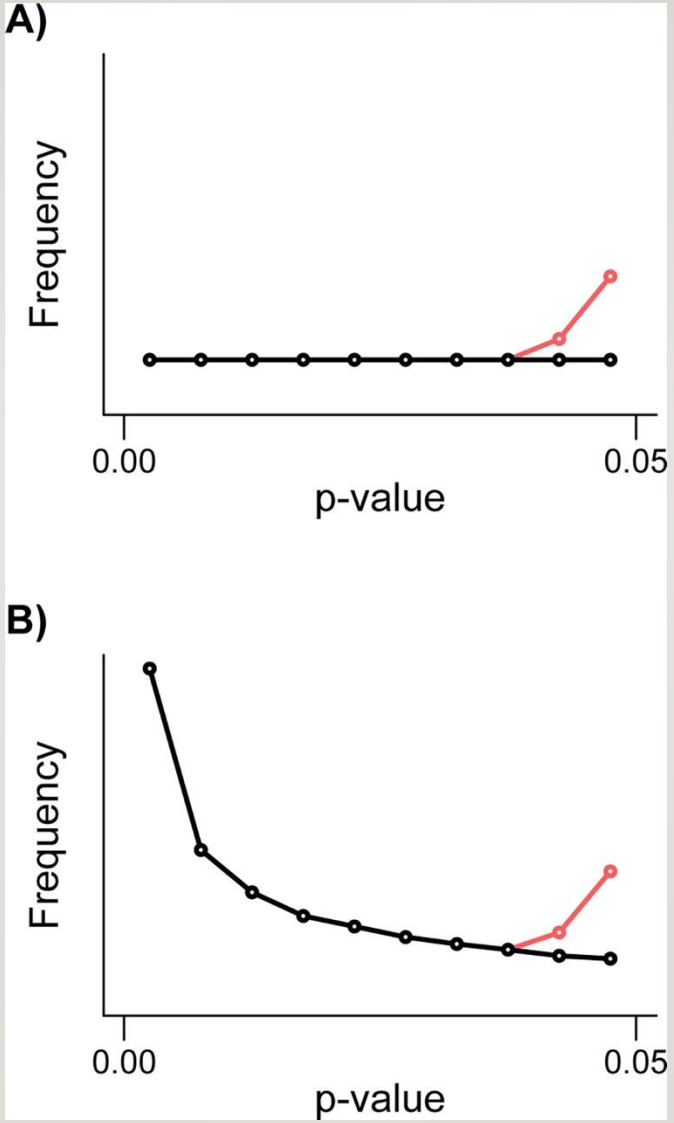

Hospital Size Cutoffs
DICON
Proposed
Number of Hospitals
0
1
2
3
4
5
0
50000
100000
150000
Yearly Patient–Days

P-HACKING AND PUBLICATION BIAS

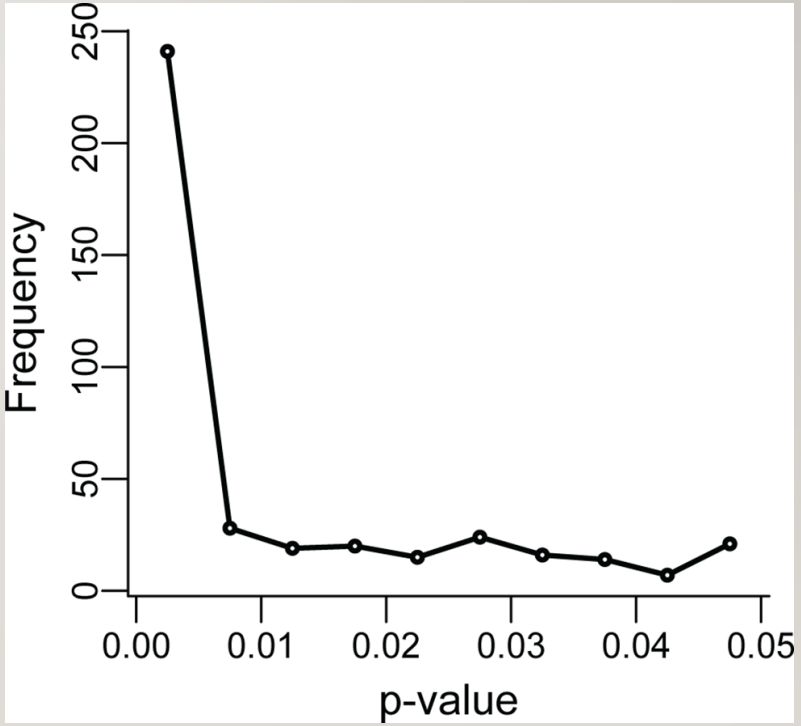
- One application of this type of visualization is detecting things like p-hacking and publication bias in a meta-analysis
- P-values should follow a theoretical distribution if they're not being influenced by researcher or publisher judgement
- But do they?
- Head ML, Holman L, Lanfear R, Kahn AT, Jennions MD (2015) The Extent and Consequences of P-Hacking in Science. *PLoS Biol* 13(3): e1002106. https://doi.org/10.1371/journal.pbio.1002106



Publication Bias



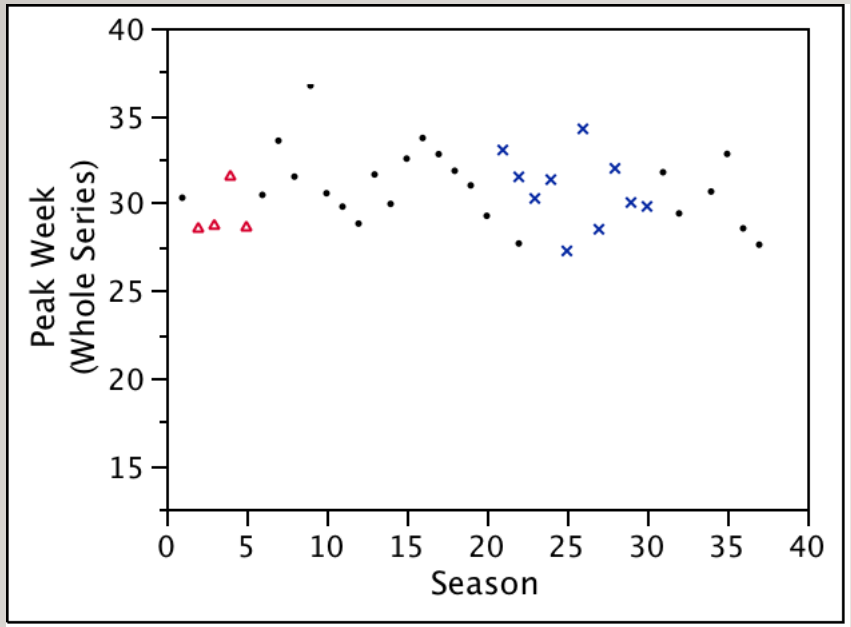
P-Hacking

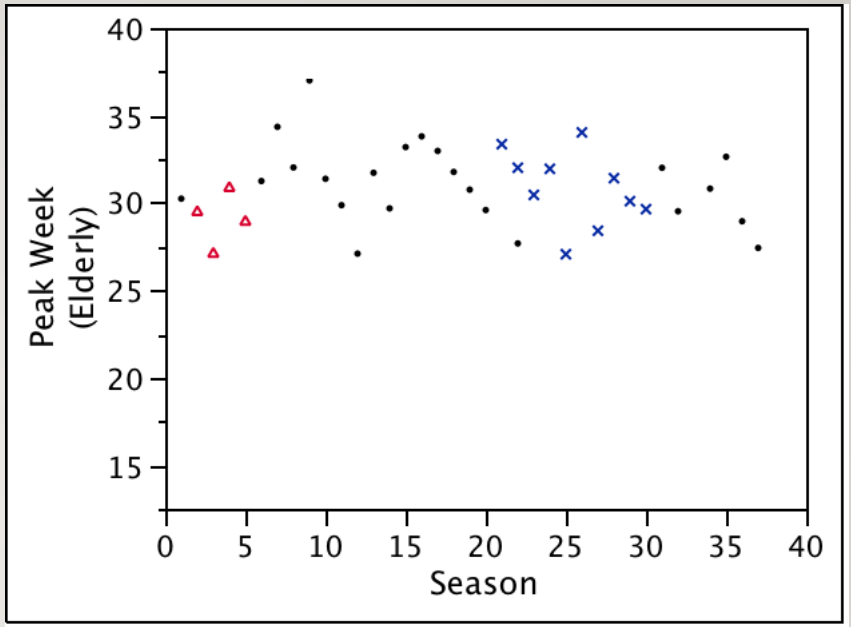


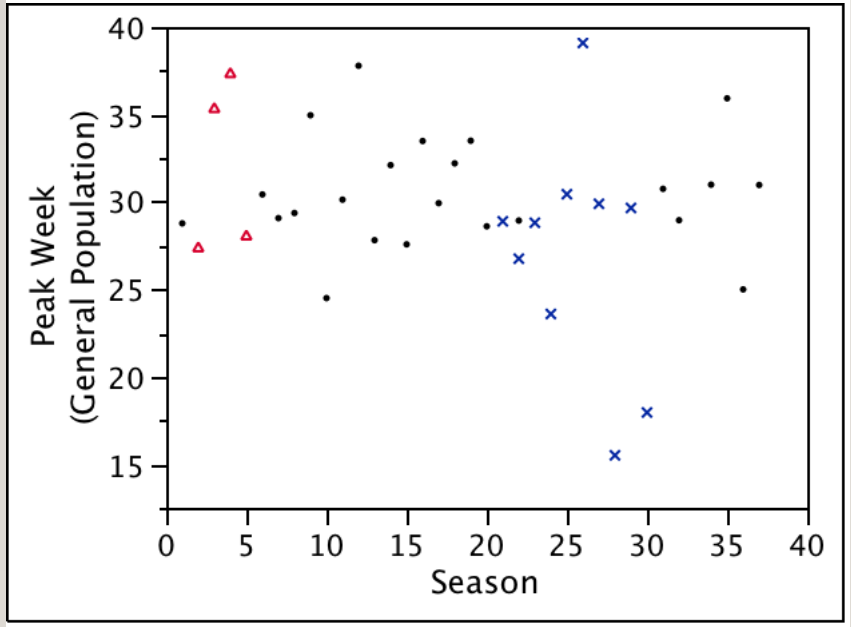
A Meta-Analysis

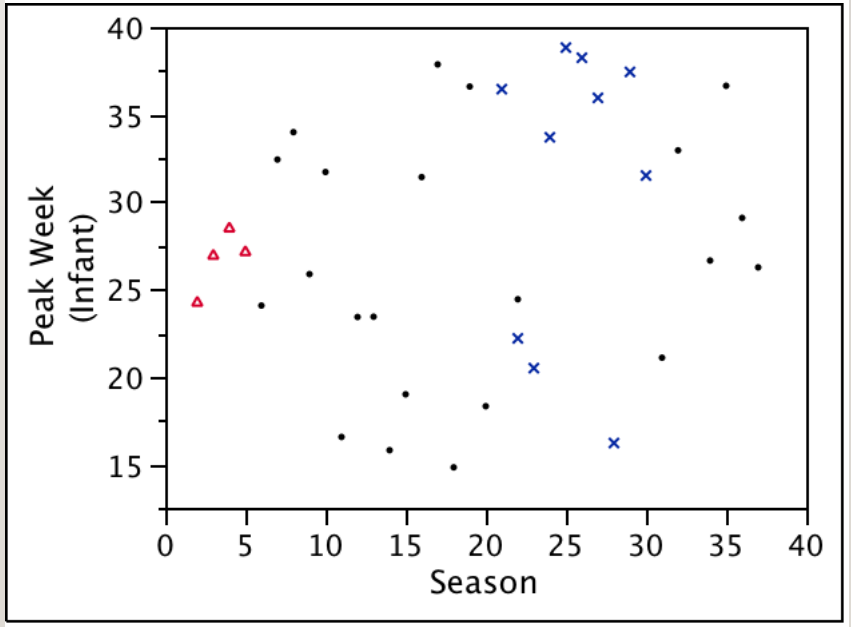
MULTIVARIATE ANALYSIS

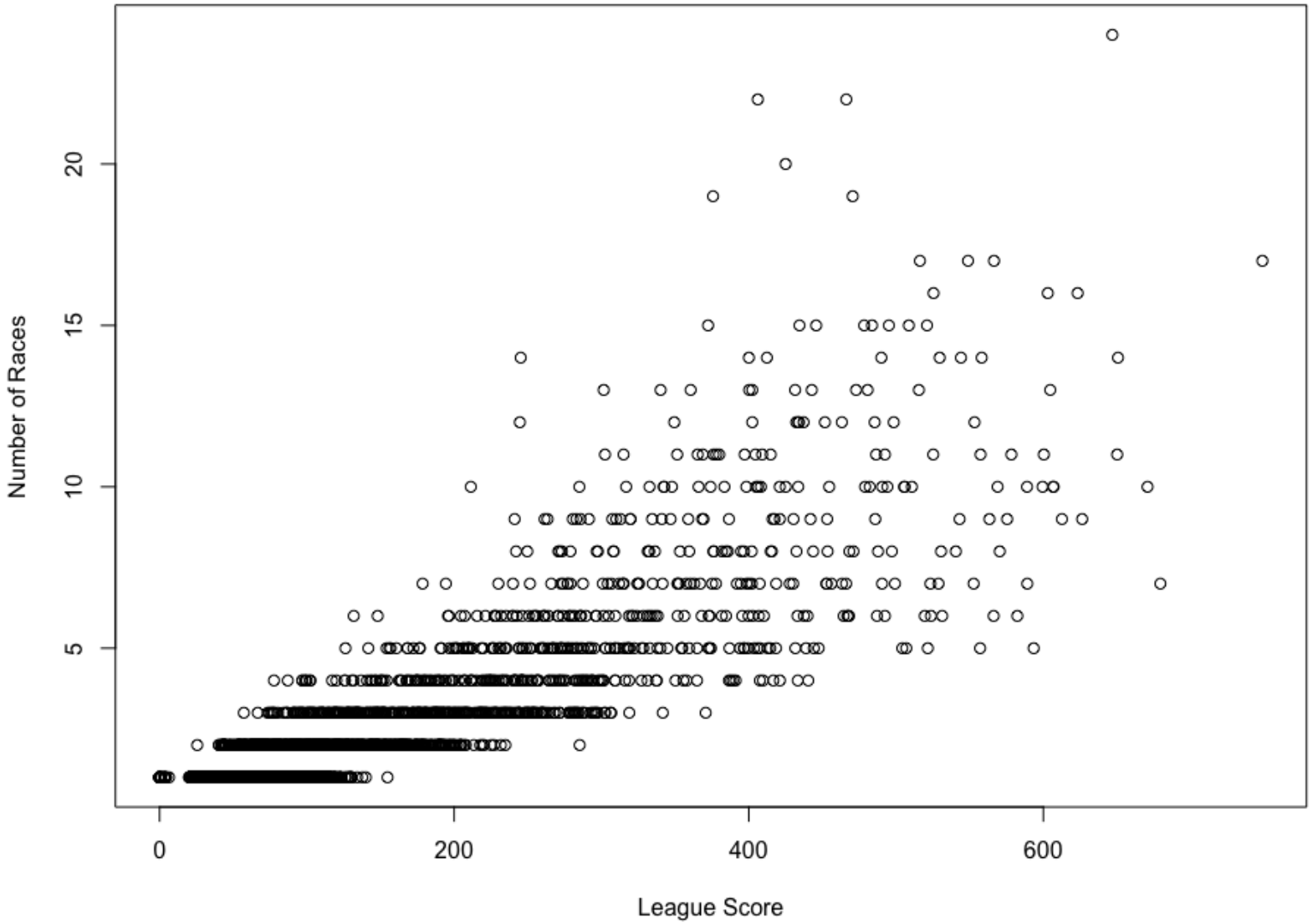
- Much of EDA is concerned with understanding the relationship between two or more variables, for a number of reasons
- These can be visualized with scatter plots, box or violin plots, and a number of other methods
- There's also quantitative methods such as correlation coefficients, or linear models of the relationships
  - Once again, to be covered in a future module

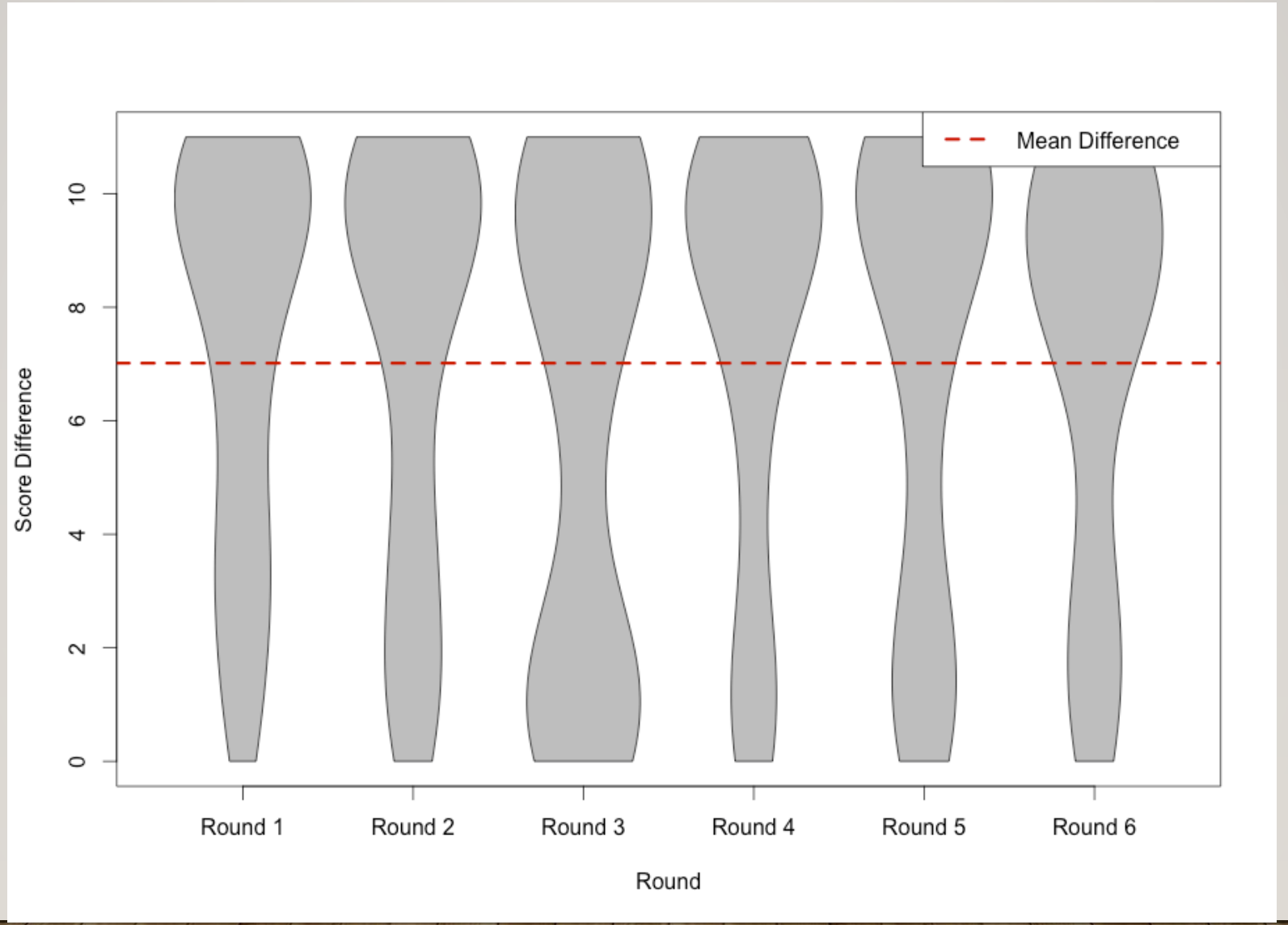

Peak Week
(Whole Series)
40
35
30
25
20
15
0
5
10
15
20
25
30
35
40
Season


Peak Week
(Elderly)
40
35
30
25
20
15
0
5
10
15
20
25
30
35
40
Season


Peak Week
(General Population)
40
35
30
25
20
15
0
5
10
15
20
25
30
35
40
Season


Peak Week
(Infant)
40
35
30
25
20
15
0
5
10
15
20
25
30
35
40
Season


Number of Races
5
10
15
20
0
200
400
600
League Score

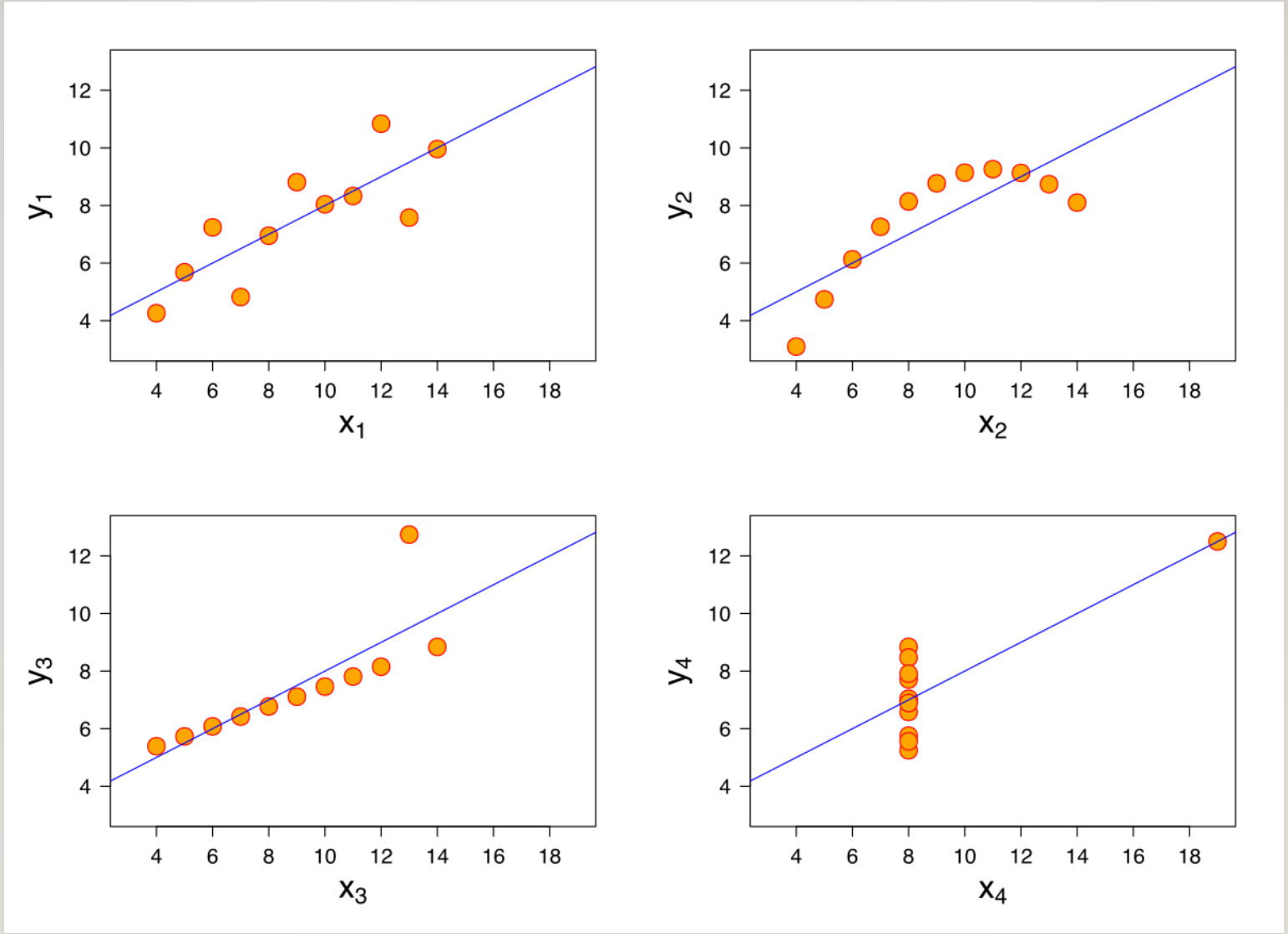

Mean Difference
Score Difference
0
2
4
6
8
10
Round 1
Round 2
Round 3
Round 4
Round 5
Round 6
Round

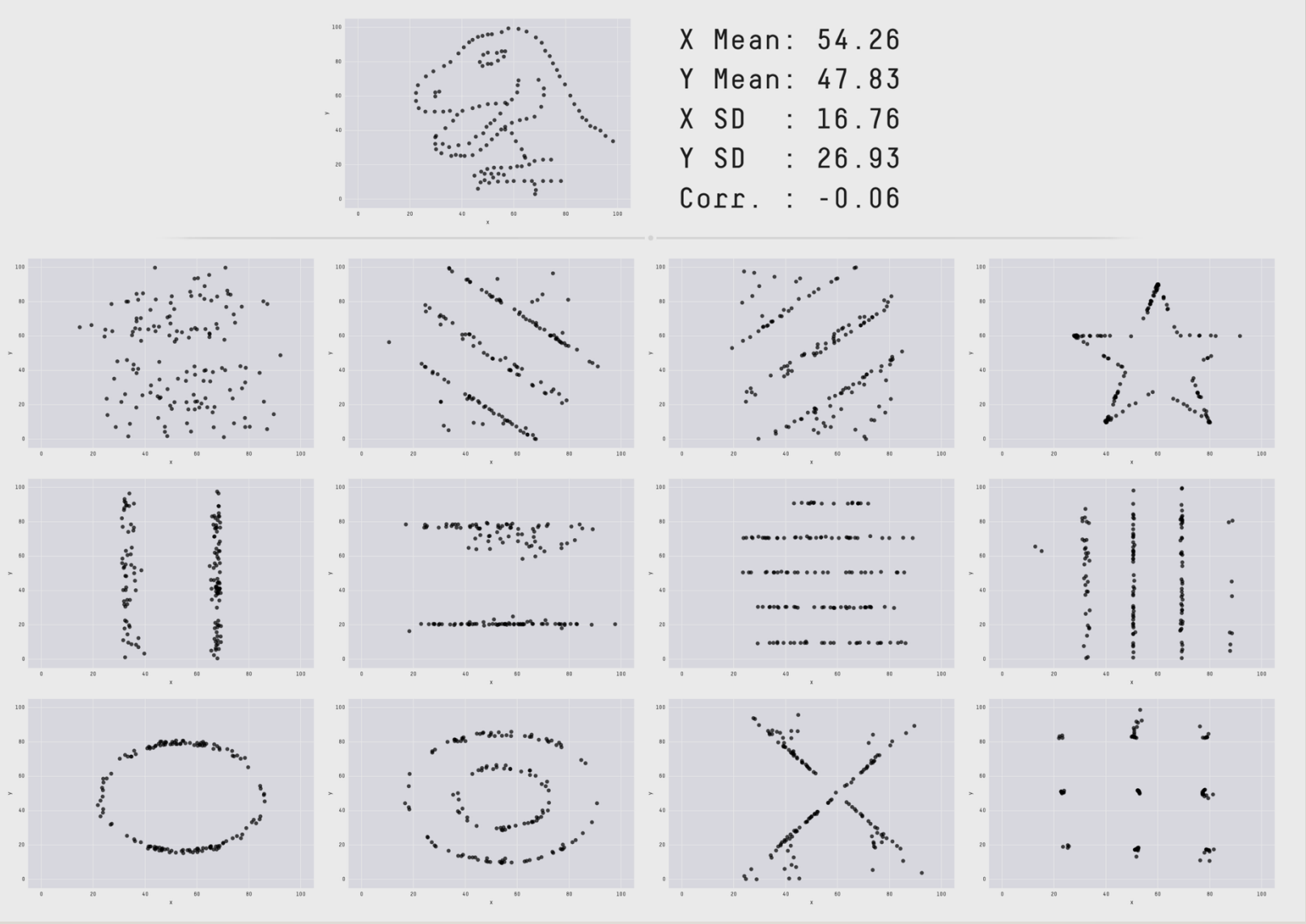
THE IMPORTANCE OF VISUALIZATION

- There is often a perception that numerical measures – means, medians, correlation coefficients, etc. are cold and precise and preferable to the fuzzy business of looking at a graph and going "That's odd…"
- Human pattern recognition is a double-edged sword – it's very good, but it's also overtuned
- Francis Anscombe designed four datasets, each with 11 points, in 1973 to explore this phenomenon

ANSCOMBE'S QUARTET

- Each dataset has:
  - An exact mean of X (9)
  - An exact variance of X (11)
  - A mean of Y to two decimal places (7.50)
  - A variance of Y of 4.125±0.003
  - A correlation between X and Y of 0.816 to three decimal places
  - A linear regression line of y = 3.00 + 0.500x to two and three decimal places respectively
  - A coefficient of determination (R^2) of 0.67 to two decimal places


y1
x1
y2
x2
y3
x3
y4
x4
4
6
8
10
12
14
16
18



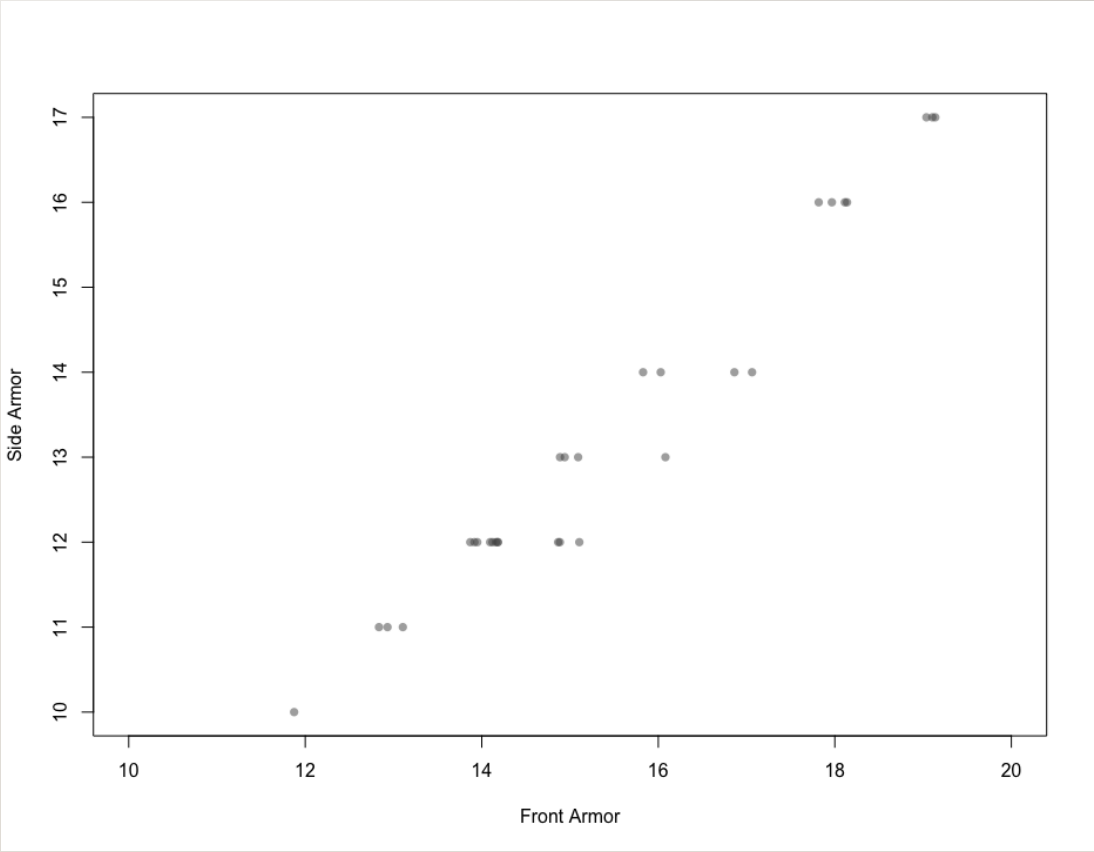
Justin Matejka and George Fitzmaurice. 2017. Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing. In Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17). Association for Computing Machinery, New York, NY, USA, 1290–1294. https://doi.org/10.1145/3025453.3025912

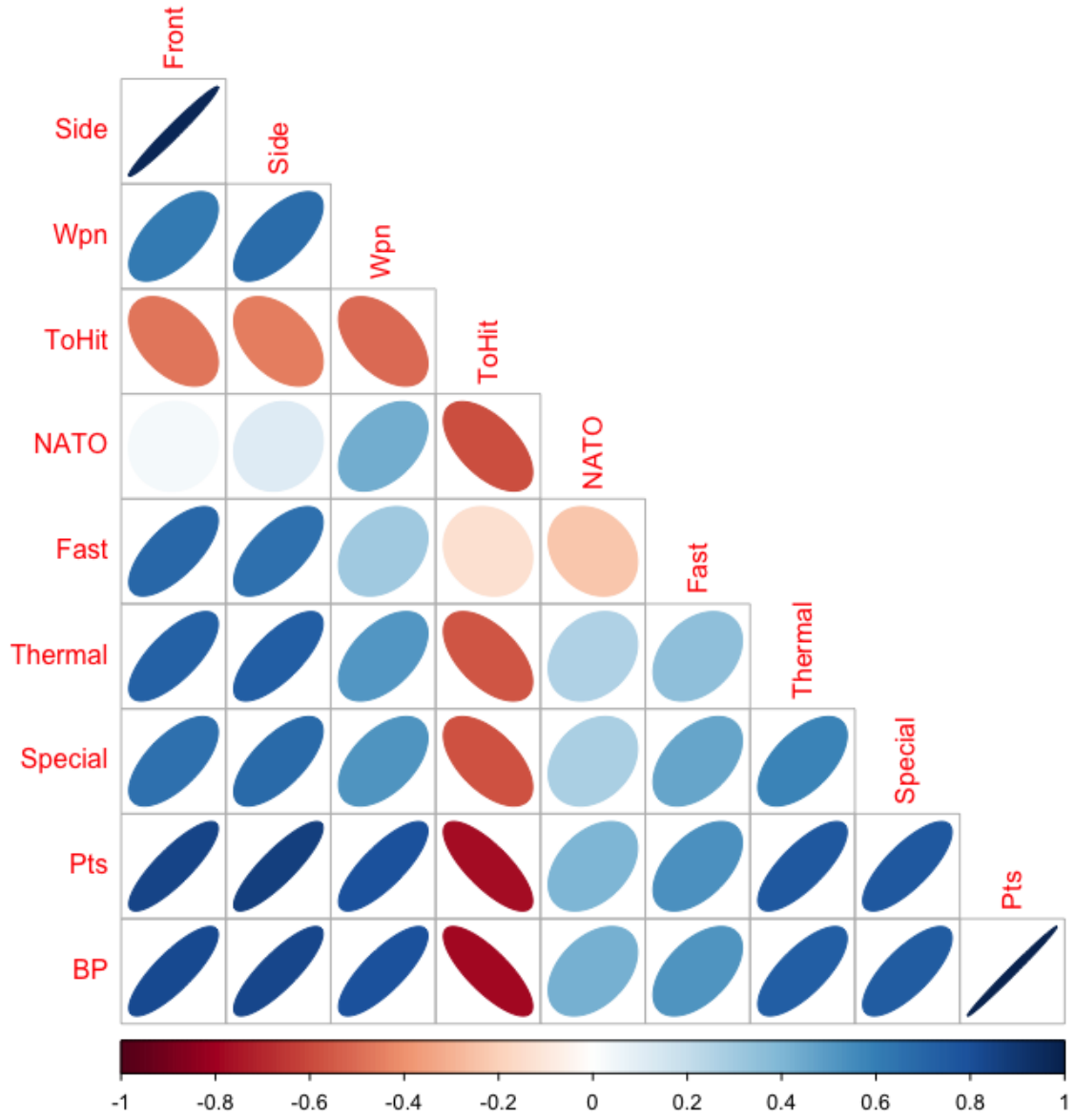
SPECIAL THINGS TO LOOK OUT FOR

- Very strong correlations
- Missing data patterns
  - Especially missing data patterns associated with another variable
- Areas of empty data
  - i.e. "empty cells"

VERY STRONG CORRELATIONS

- Very strong correlations suggest that one variable very well predicts another
- If this is what you *want*, good
- If you think you're going to include both of these variables in an analysis, this is bad, as you're going to add variance for no gain in information




Front
Side
Wpn
ToHit
NATO
Fast
Thermal
Special
Pts
BP
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
0.8
1

COMPLETE SEPARATION

- What can happen in this case is that one variable completely and perfectly predicts another
- This is called "complete separation"
  - There's also quasi-complete separation, which is you're dangerously close to this
- This is a problem, because you're likely not explaining something, but rather statistically representing a tautology

MISSING DATA PATTERNS

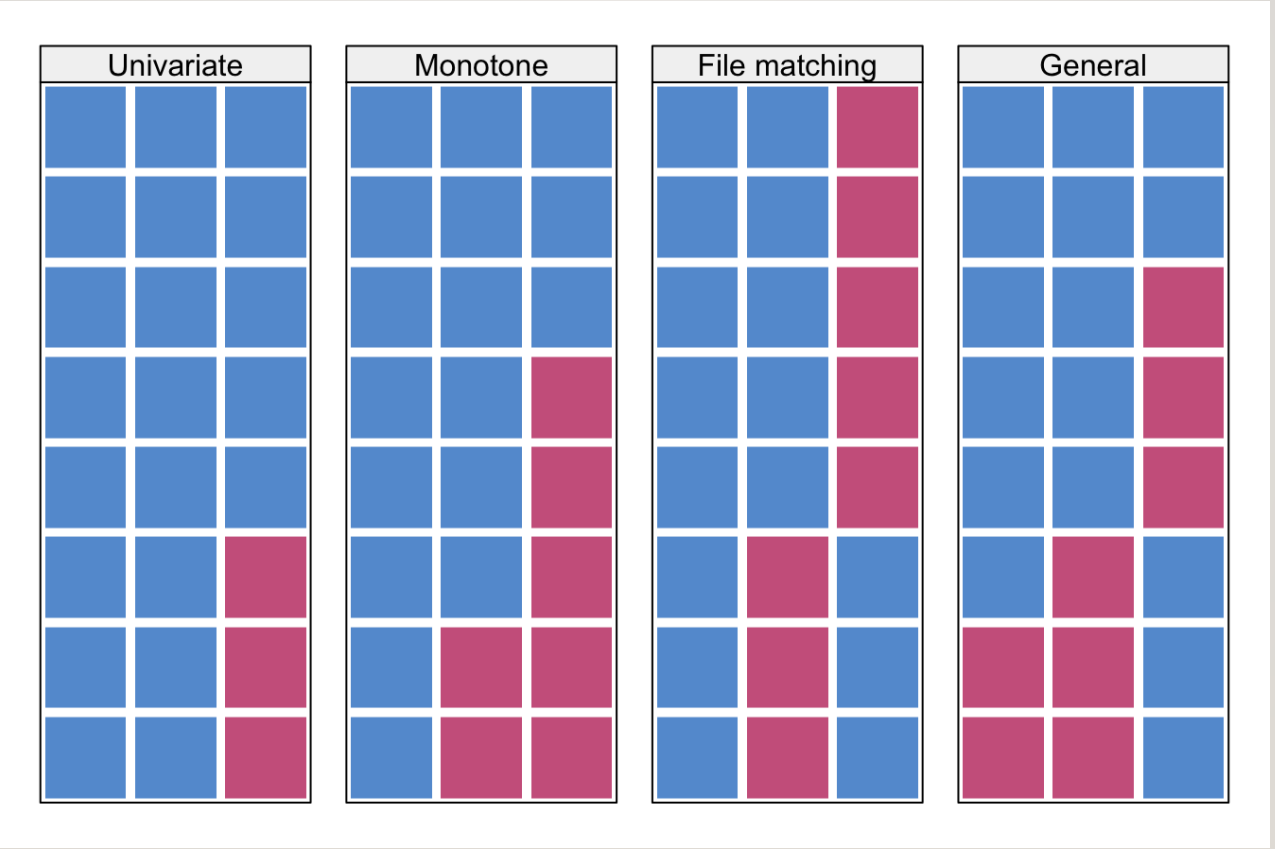


Figure by Stefan van Burren

EMPTY CELLS

- One of the underlying assumption of many of the methods we use is called "Positivity"
- In the observed data, there must be a non-zero probability of every individual or unit receiving any level of the treatment or exposure variable
- You take a sample of students in Pullman, and one of the variables you ask about is whether or not they drive a stick shift car
- Zero students say yes
- Is this a violation of the positivity assumption?

AN EXAMPLE

- You have been hired by the United States Navy to study non-combat related injuries in sailors
- You have data from 1967 to 2010 (the next 15 years remain classified)
- You have been given a number of variables: Age, what type of ship the sailor serves on, their rank and job, gender, physical fitness scores, height, weight, etc.
- Can anyone guess the problem?

POSITIVITY PROBLEMS

- Merely not having someone in a cell is not a problem
  - Again, think stochastically – how did someone end up in your sample, and what alternative samples might there be
  - Rare is not impossible
  - But impossible is bad
- These questions come up a lot in social epidemiology and related fields, and can get somewhat philosophical
  - Is a minoritized population that is not found in the highest income neighborhoods, while the majority population is not found in the lowest income neighborhoods (with overlap in the middle) a positivity violation or not?

QUESTIONS?

THE FINAL

- From the Syllabus: "You will select a topic of interest to you, based on your own work or as part of the material we have covered over the course of the semester. The purpose of this project is to identify a problem – including prior research in the area and outline how a statistical approach may aid you in answering it."
- What does this mean?
  - This is your chance to apply what you've learned in this class to a problem you're interested in
    - It should be biomedical, but it need not be the project you are currently actively working on, your dissertation topic, etc.
  - Essentially, what I'm looking for is a brief review of the literature to introduce the topic, followed by you considering the statistical approach to your problem up until you have to *do* the statistics
    - What is your source population? Do you have a target population? A sampling frame? What considerations would you put into sampling? Are there major threats to your approach – bias in sampling? Logistical difficulties? Possibly nonpositivity?
    - What type of hypothesis are you testing, or is this "just" estimation
      - Not hypothesis testing and "What's your hypothesis" in the colloquial sense can be different

FORMATTING AND OTHER GUIDELINES

- Your final paper should be 5 pages
- This is a maximum, not a minimum
- I'm not going assign specific formatting guidelines, but don't be clever
- A broadly applicable piece of advice in academia for grant writing, getting papers through peer review, etc.: Don't annoy your reviewers
- CITE YOUR SOURCES
  - Again, there is no specific citation format, so pick one
  - References do not count against your page limit

WHAT IS HYPOTHESIS TESTING?

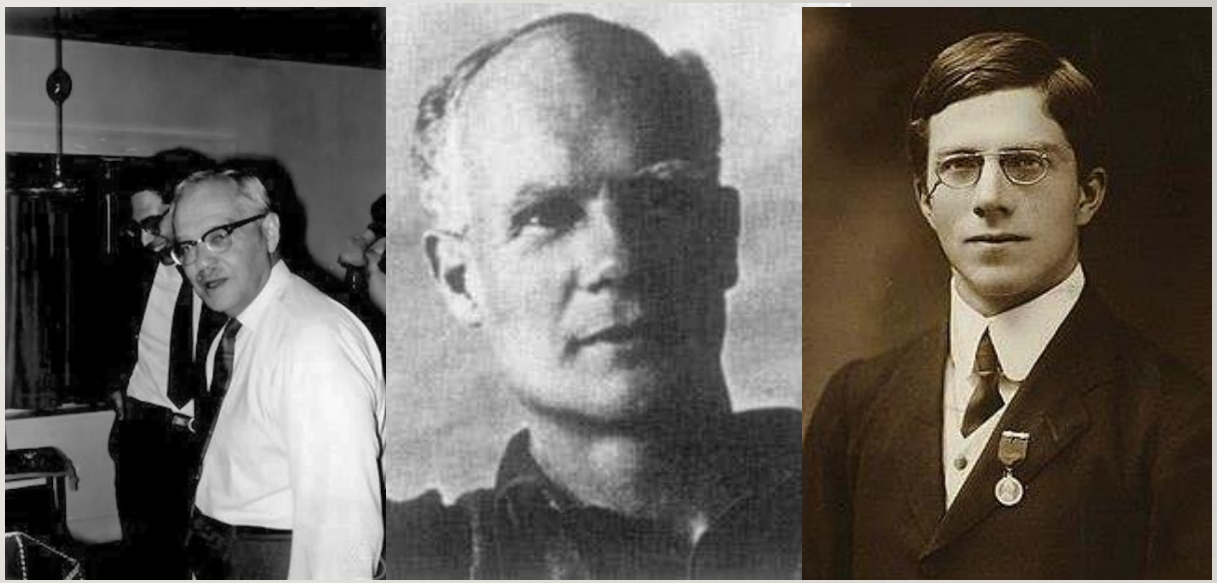
- Hypothesis testing is a formal procedure for interrogating how well our ideas about the real world cohere with with our data
- It is the process of determining how well a "null" hypothesis about a *population* quantity fits a *sample* of the data
  - This bit is important – while we are using our sample data, we are asking a question about the population
- Estimation can tell you how big an effect is, while hypothesis testing tells you if there's any effect at all

WHAT IS A *STATISTICAL* HYPOTHESIS?

- A statistical hypothesis is a precise, quantitative statement about a population, which is then evaluated to see if the sampled data supports it
- In hypothesis testing, we first consider a "null hypothesis"
  - This is *often* but need not always be a "nil hypothesis" (i.e. a statement that there is no effect)
  - Usually, this is denoted H_0
- There is also an "alternative hypothesis"
  - This is usually denoted H_A
  - This is a mutually exclusive alternative to the null hypothesis

WHERE THIS GETS MESSY AND COMPLICATED

- Hypothesis testing is often called "Neyman-Pearson Hypothesis Testing" after two collaborators who developed much of the theory
- Ronald Fischer *also* developed much of the theory, and popularized the use of the phrase "significance test"
- They disagreed on whether or not you needed an alternative hypothesis
- This debate was not settled when Fisher died in 1962



Neyman, Pearson and Fisher

FISHER'S APPROACH

1. Set up your null hypothesis
2. Report the level of significance you find (we'll discuss what that means soon)
3. If your result is not significant, you can draw no conclusions or make no decisions without more data

Fisher advised only using this when little is known about your problem – he started life as a Bayesian and asserted that these were sort of provisional findings that ignored wider data.

This is sometimes called "Null Hypothesis Testing" for obvious reasons, and is often most concerned with asking if the observed data are consistent with having arisen from chance

NEYMAN-PEARSON APPROACH

- Set up a null and alternative hypothesis, and determine acceptable rates of alpha (the rate of fall positives) and beta (the rate of false negatives) based on considerations of costs-benefit, etc.
  - Even relatively cold hypothesis testing involves subjectivity
- If the data falls into the rejection region of the null hypothesis, accept the alternative hypothesis
  - This does not mean you think it is true, just that you'll act on it as if it is
- Here you *need* disjoint hypotheses, and you need to be able to meaningfully make a cost-benefit argument for alpha and beta

THE MODERN METHOD

- We just sort of…crash the two into each other
  - We start using Fisher's p-value concept as a substitute for Neyman-Pearson's more complicated likelihood ratio test in many cases
- This is a messy hybrid that has a whole mess of hidden assumptions and caveats
- Neyman and Pearson arguably had the much more rigorous form, but what is often taught is much like Fisher's conception
  - This includes things like primarily being concerned with an alternative hypothesis that is "Not The Null Hypothesis" rather than an actual alternative hypothesis
  - Neyman-Pearson testing allows for the estimation of both Type I and Type II error
  - But we've largely abdicated any meaningful discussion about our choices of hypothesis and alpha and beta to field-specific conventions

TYPE I AND TYPE II ERROR

- Type I error: This type of error occurs when we reject a null hypothesis that is actually true
- Type II error: We fail to reject a null hypothesis that is actually false
- Because we are often looking for an effect in a population, these are referred to as "False Positive" (i.e. we said an effect existed when there wasn't one) and "False Negative" (we said there wasn't an effect when there was) but conceptually, they can be broader than that

IS THIS A GOOD IDEA?

- "We are quite in danger of sending highly trained and highly intelligent young men out into the world with tables of erroneous numbers under their arms, and with a dense fog in the place where their brains ought to be. In this century, of course, they will be working on guided missiles and advising the medical profession on the control of disease, and there is no limit to the extent to which they could impede every sort of national effort." – R.A. Fisher
- Fisher was an expertimentalist and evolutionary biologist in addition to being a statistician, and was concerned (rightly) with how very clean mathematical notions of probability intersected with the realities of empirical science
- Before we get too carried away, he was also a eugenist

THE END RESULT

- The end result of this is confused students (as predicted)
- Hypothesis testing is massively powerful, often quite misunderstood, and in practice we somewhat limit ourselves to well understood hypothesis tests in many cases
- The *bulk* of hypothesis testing involves a null hypothesis, and an alternative hypothesis that is simply "not the null"
- Quite often, H_O is that a particular sample distribution is indistinguishable from one arising from chance, or for the estimates of many effects, that the effect is zero
- This is not always the case. Likelihood ratio tests crop up in model selection, which are testing disjoint hypotheses that one model is better supported than another

LADY TASTING TEA

- Many of you may know that the British have particularly strong feelings about the combination of milk and tea
- Namely, that you add milk to tea, not the other way around
- Does anyone know *why?*

LADY TASTING TEA

- Dr. Blanche Muriel Bristol, who worked with Fisher at the Rothamsted Experimental Station, claimed to be able to distinguish the taste of tea added to milk vs. milk added to tea
- Fisher thought this was nonsense
- A third colleague, William Roach, suggested they test her
  - She would, incidentally, marry William four years later
- They devised an experiment where four cups of each type were presented to her, randomly

WHAT'S THE NULL HYPOTHESIS?

THE EXPERIMENTAL SETUP

- There are 70 possible combinations of results:

Successes	Combinations	Number of Combinations*
0	XXXX	1
1	XXX✓,XX✓X,X✓XX, ✓XXX	16
2	XX✓✓, X✓X✓, X✓✓X, ✓X✓X, ✓✓XX, ✓XX✓	36
3	X✓✓✓, ✓X✓✓, ✓✓X✓, ✓✓✓X	16
4	✓✓✓✓	1

*This follows what's called a hypergeometric distribution

HOW DO WE EVALUATE THE NULL HYPOTHESIS?

- Fisher thought that a 1-in-20 chance of mistakenly rejecting a null hypothesis that was true was acceptable
- Putting it in Neyman-Pearson terms, alpha = 0.05
- This turns into a p-value of 0.05 as a cutoff
- Would anyone like to hazard a guess as to how he determined this?


VIBES

BACK TO TEA…

- Out of the 70 possible combinations, only one combination (all four cups correctly identified) had a less than 5% chance of occurring by chance (having a 1.4% chance)
- This meant that Fisher would only reject the null hypothesis that Dr. Bristol could not detect a difference if she correctly identified all four cups
- She did so



THE IMPORTANCE OF ASKING THE RIGHT QUESTION

- It should be obvious now that the details of your null hypothesis is critically important
- "Greater than", "Less than", and "Different from" mean different things, and have mathematical consequences
- You want to be sure your statistical hypothesis is in support of the scientific question you think you're asking
- Importantly, you want to know *what your statistical hypothesis is*
  - "Here's my data, I need a p-value" will kill the soul of any statistician you work with

QUANTIFYING UNCERTAINTY

- All of this is about quantifying uncertainty
- How much does our data support a particular argument about something that cannot be measured in the population, but can be measured in our sample?
- We have tools to assess this:
  - Estimation: Confidence Intervals
  - Hypothesis Testing: P-values
  - Bayesian Methods: Posterior Distributions
- The first two are two sides of the same coin
- The third we'll discuss in a later class

SIGNIFICANCE LEVEL

- Looping back, your significance value (alpha) is the probability threshold that defines the binary decision as to whether or not something is "statistically significant"

$$\alpha = \Pr(reject\ Null \mid Null = True)$$

- We say a result is statistically significant when p < alpha
- Alpha is fixed to the amount of uncertainty we're okay with
  - This is most often 0.05 – though as mentioned previously, this is largely arbitrary
  - For some fields, it's different. GWAS studies, for example, demand much smaller values for alpha
- Your confidence level is 1-alpha
  - Hence 95% confidence intervals

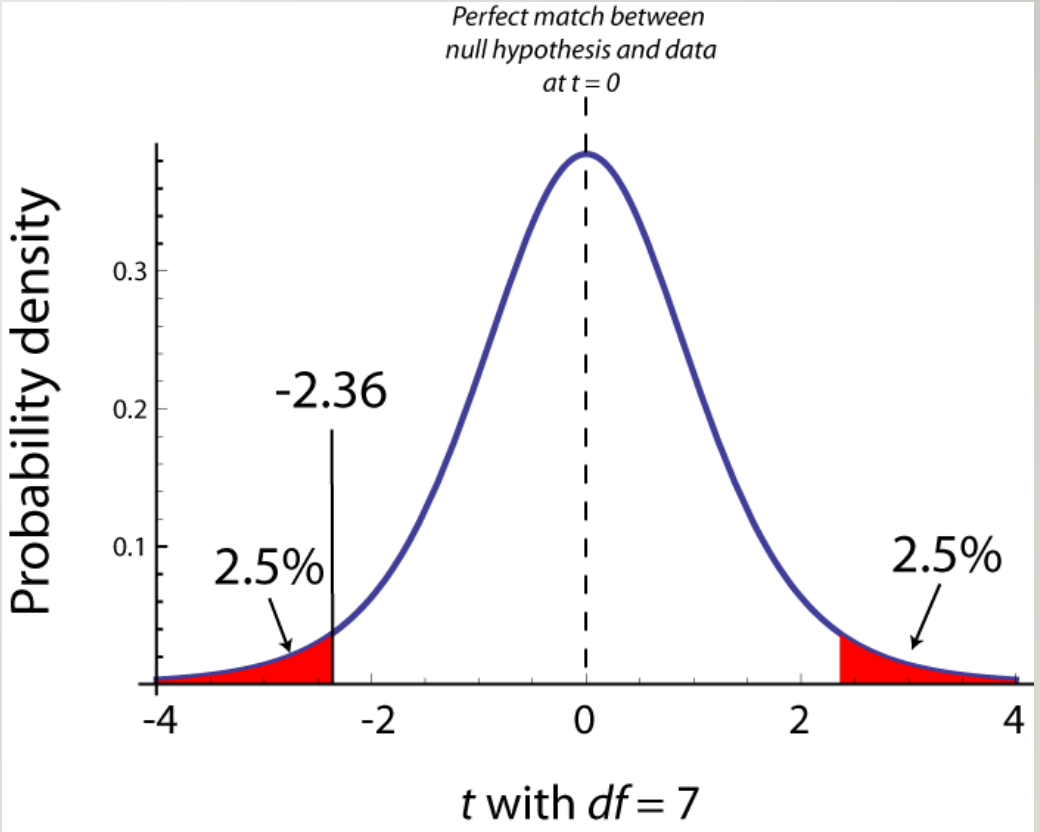
P-VALUES

- Think stochastically – your data is one of *many* potential random draws from your population
- The p-value, which we calculate from a statistical test, described how likely your particular data would have arisen if the null hypothesis was true
  - i.e. Pr(Data | Null Hypothesis)
- A p-value of 0.17 means there's a 17% chance your data could have occurred under the null hypothesis
- If $p < 0.05$, there is a less than 5% chance
- This is *not* the probability that the alternative hypothesis is "true", the probability you're right, etc.

CRITICAL VALUES AND THE NULL DISTRIBUTION

- Null distributions are used to quantify the uncertainty of your hypothesis test.
- Null distributions are classically the distribution of a test statistic when the null hypothesis is true.
- Student's *t-distribution:*

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$



ON SIGNIFICANCE

- Significance. Is. Arbitrary.
- Yet we treat it like it's very, very important
- The hurdle between getting a study published when $p=0.051$ and $p=0.049$ is massive, and yet if I told you I had a drug that could cure cancer, but it wasn't significant at $p=0.051$, you'd still be interested
- Don't fall into the rote tendency to just check if $p<0.05$ and treat that as the sole arbiter of if something is interesting or not

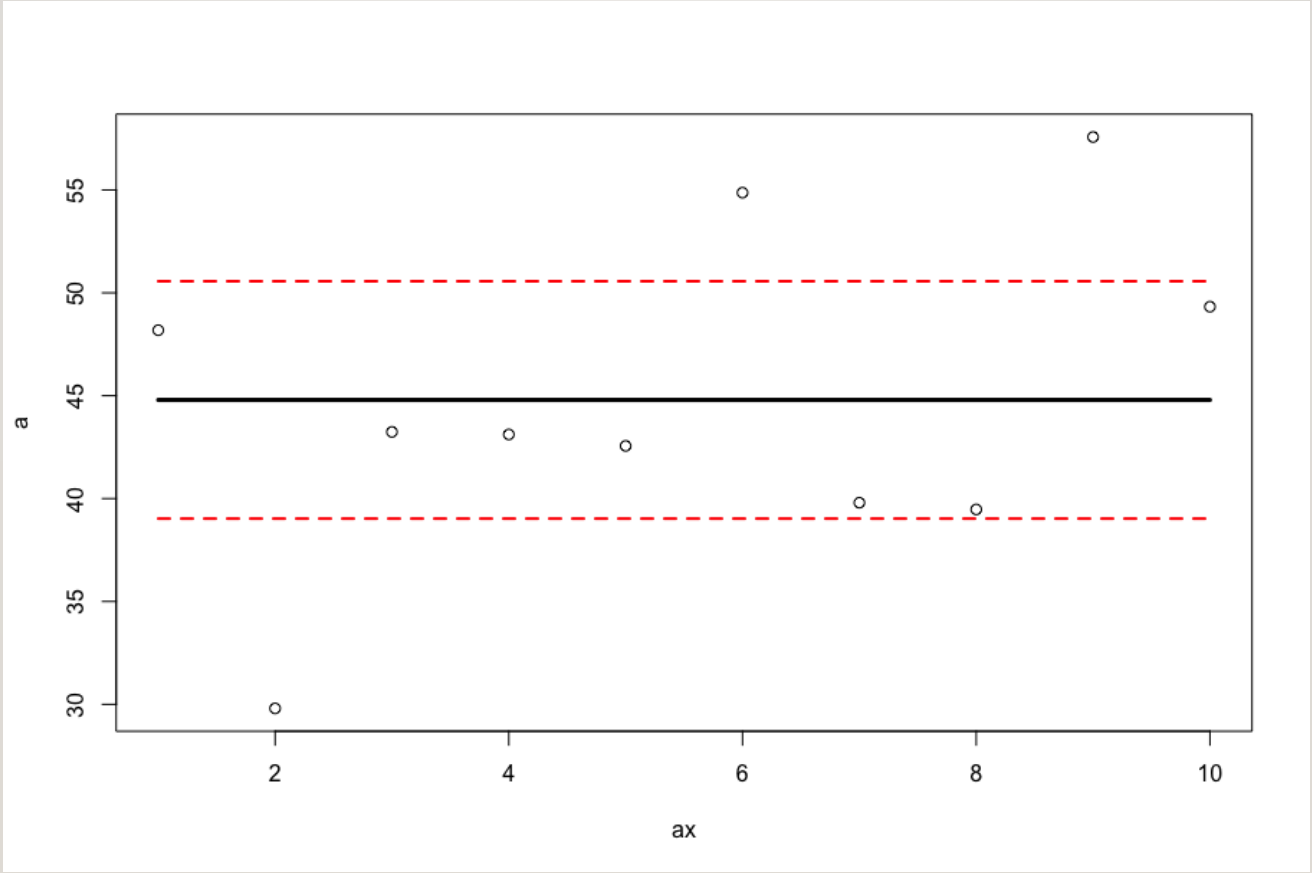
CONFIDENCE INTERVALS

- Confidence intervals are how we measure the uncertainty of estimates
  - This can be a proportion, a variance, the difference between two population means, a correlation coefficient, etc.
- All estimates have sampling distributions and standard errors
- These are most often already determined for us and baked into our statistical tools
- Typically, a 95% confidence interval is your *estimate*±1.96**standard error*
- This is the range under which, under repeated sampling, 95% of your estimates would fall

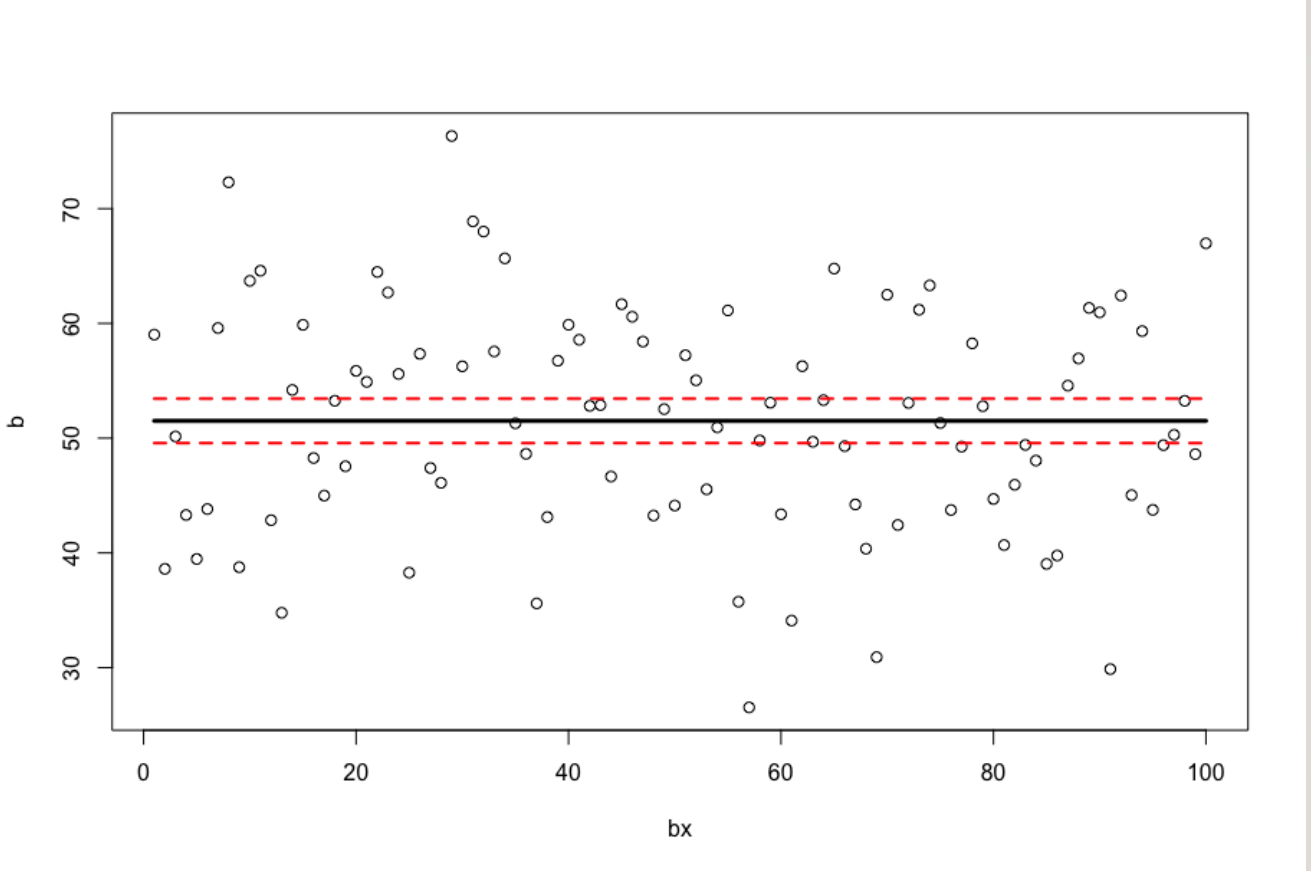
REMINDER…

- A p-value or confidence interval is a measure of *uncertainty*, not of variability
- We become more certain, for example, of our estimate
- But there may still be variability, and that we're just fleshing out if we, for example, increase our sample size

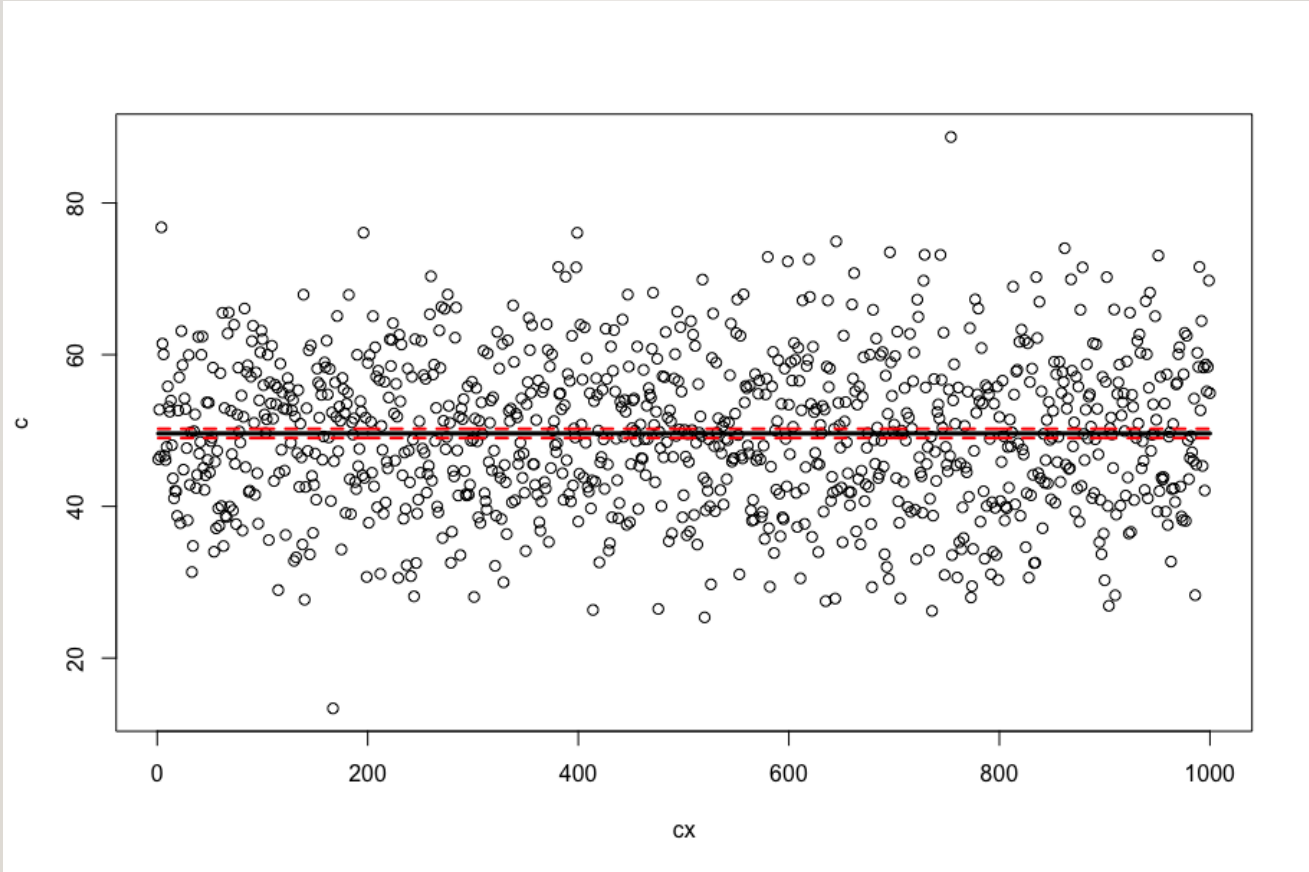
10 SAMPLES



100 SAMPLES



1000 SAMPLES



EFFECT SIZE VS. STATISTICAL SIGNIFICANCE

- Statistical significance tells you about your data as compared to some null hypothesis
- It does not tell you if an effect size (correlation coefficient, different in means, etc.) is *important* in the real world

$$r = 0.80$$

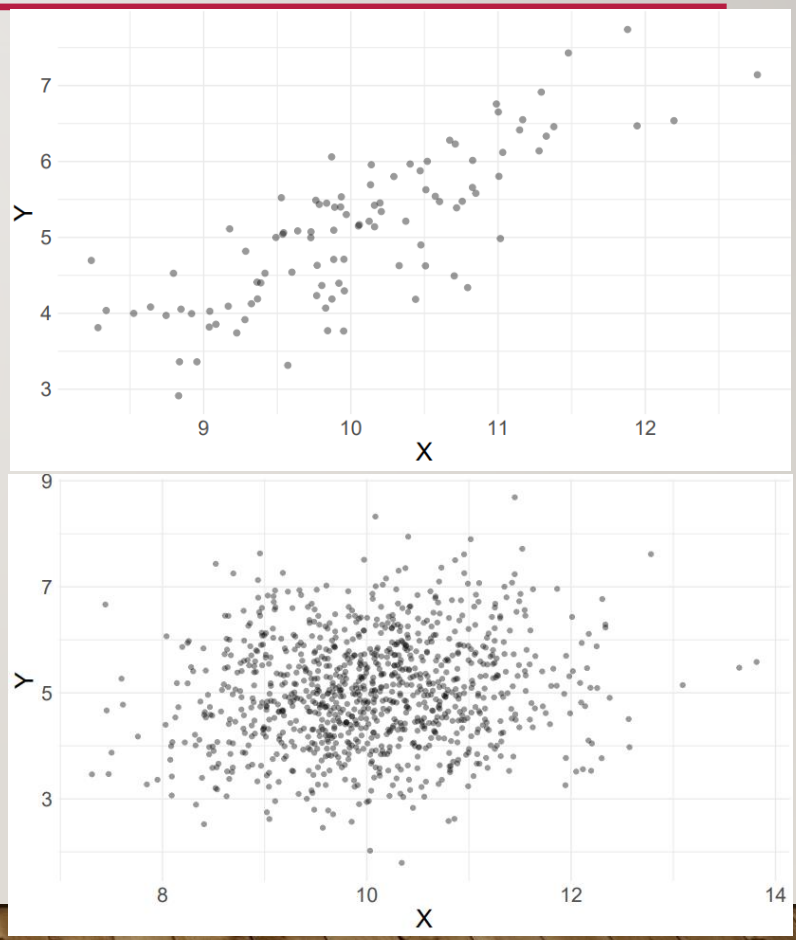
$$[0.72, 0.86]$$

$$P < 2.2 \times 10^{-16}$$

$$r = 0.14$$

$$[0.080, 0.20]$$

$$P < 7.296 \times 10^{-}6$$



Courtesy of Erin Clancey

SIGNIFICANT =/= IMPORTANT

- Statistical significance tells us how confident we can be in rejecting the null hypothesis
- What it doesn't tell us if that is important in a clinical, biological, ecological, etc. sense
- What is important depends on *context*
- It's good practice to present as much information as possible
  - Don't just say "Significant or Not"
  - Avoid "Significance Stars"
  - Present the actual p-value
  - Ideally, present the effect with the confidence interval
    - Some journals, like *Epidemiology*, particularly frown on p-values for some of the problems I've discussed earlier
  - This is also important for reproducibility, meta-analysis, etc.